Heapsort

COMS10018 - Algorithms

Dr Christian Konrad

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Data Structures

- *Data storage format that allows for efficient access and modification*
- Building block of many efficient algorithms
- For example, an array is a data structure

Priority Queues

Priority Queue:
Data structure that allows the following operations:

- Create(.): Create data structure given a set of data items
- Extract-Max(.): Remove the maximum element from the data structure and return it
- *others...*

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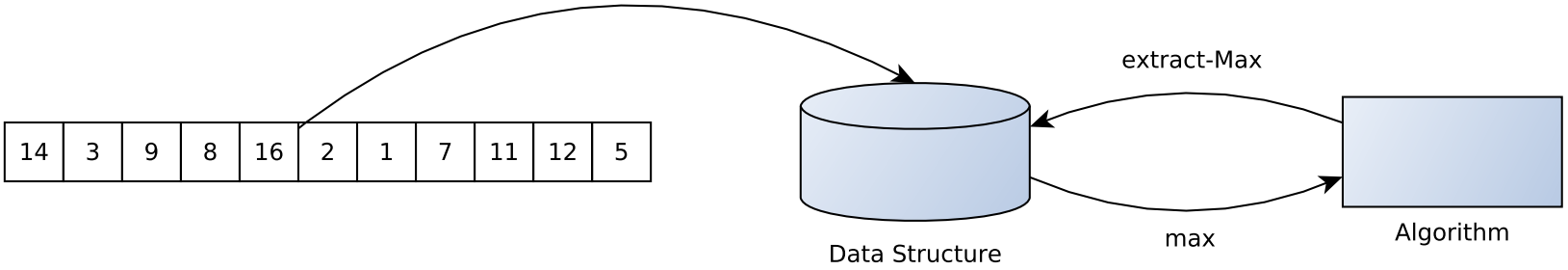
Sorting using a Priority Queue



From Array to Tree

Interpretation of an Array as a Complete Binary Tree

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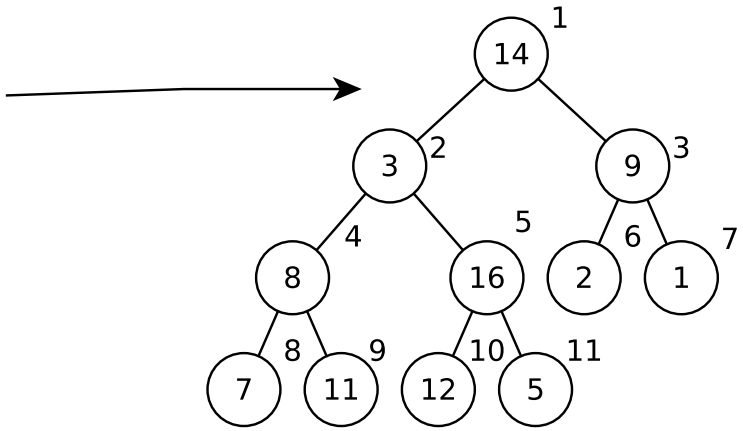
14	3	9	8	16	2	1	7	11	12	5



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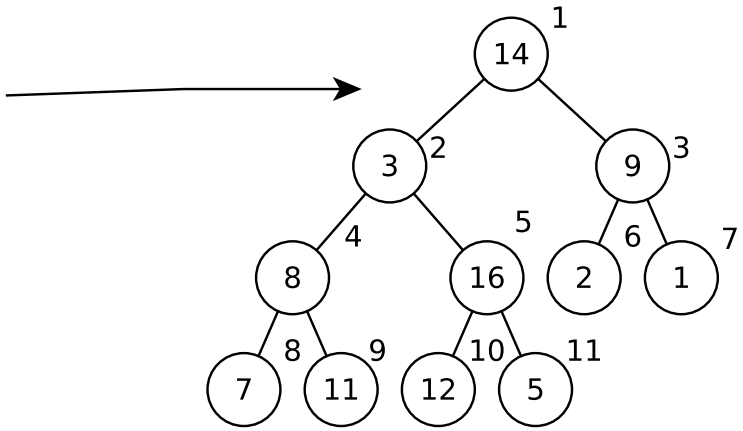

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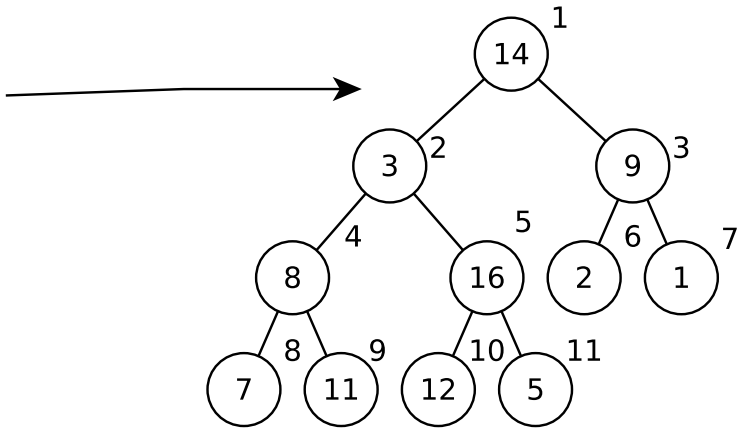

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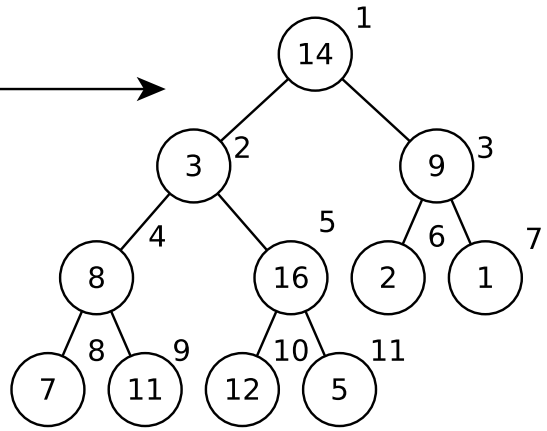

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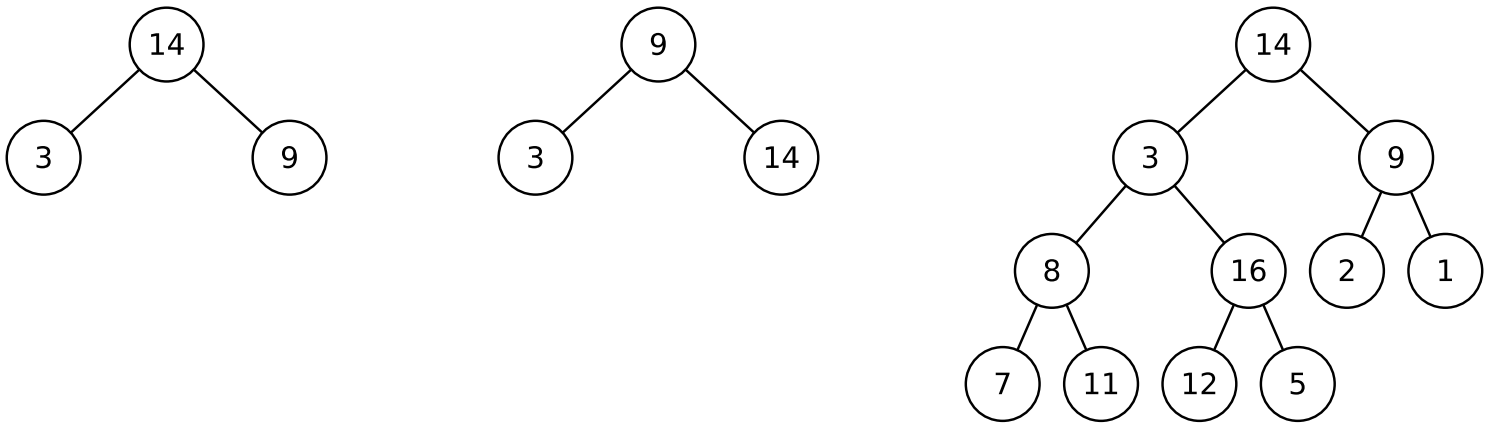
- Parent of i: $\lfloor i/2 \rfloor$
- Left/Right Child of i: $2i$ and $2i + 1$

Heap Property

The Heap Property

Key of nodes larger than keys of their children



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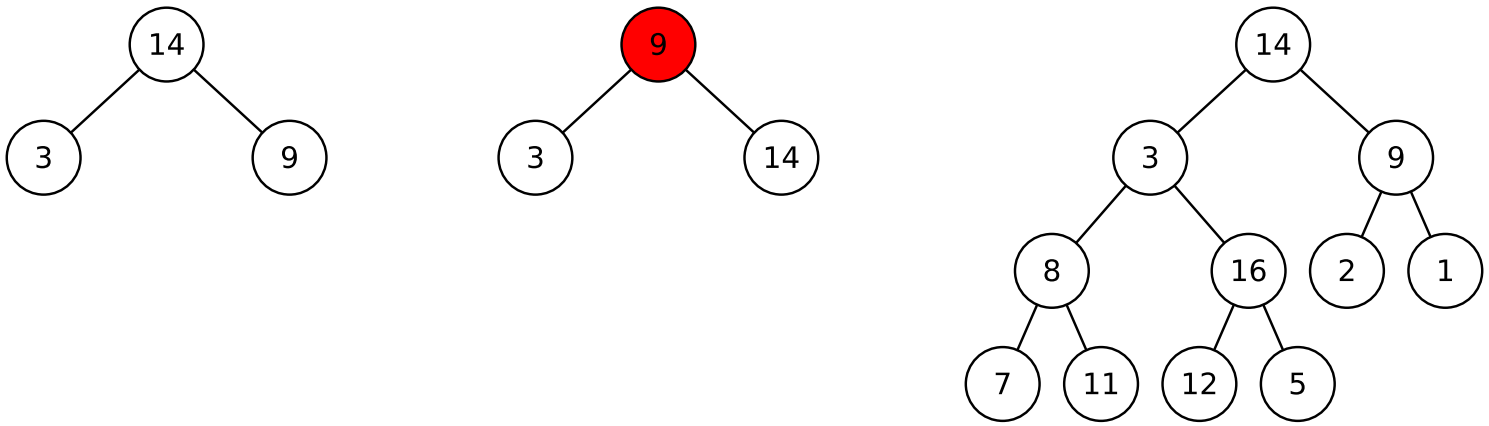

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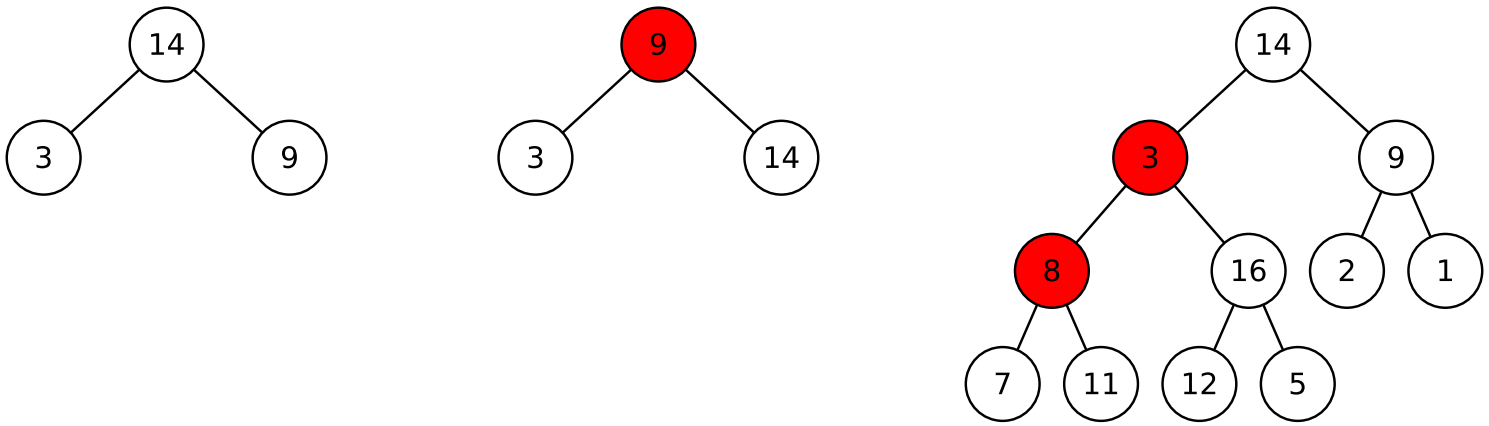

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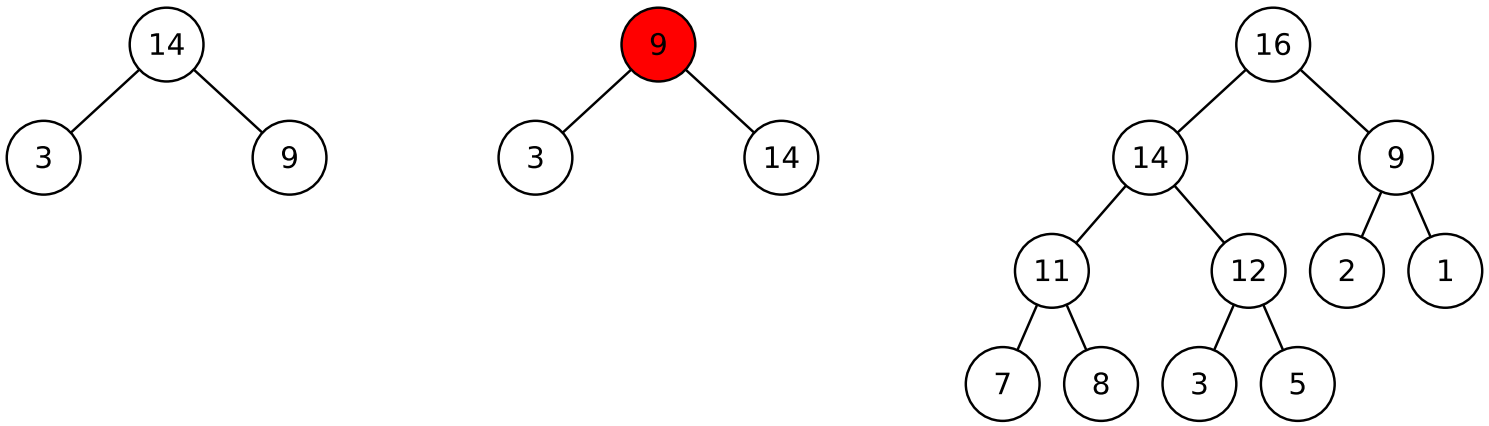

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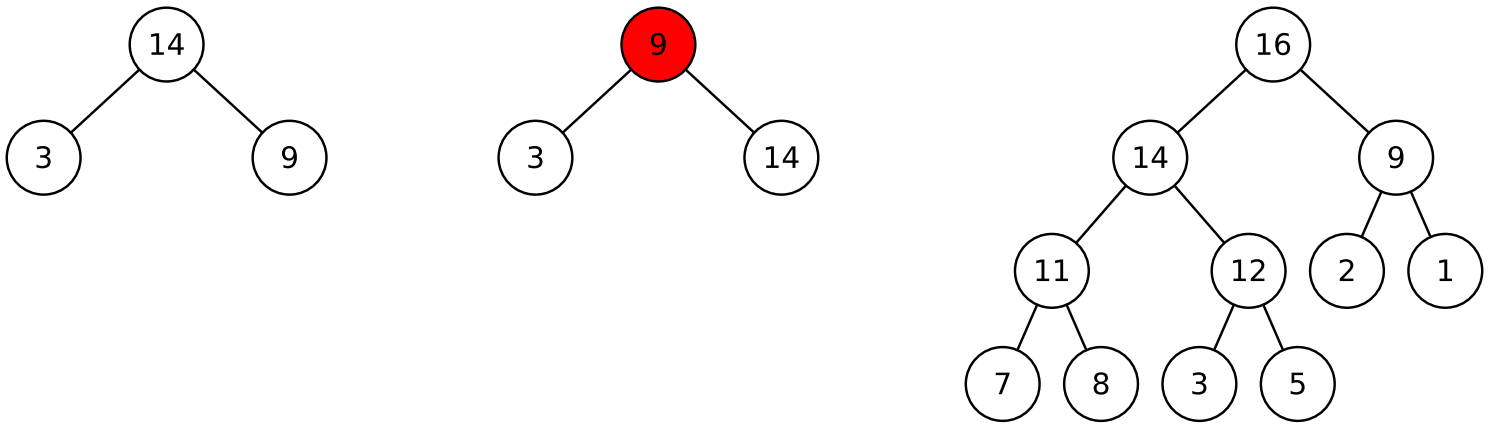

Heap Property $\rightarrow$ Maximum at root
Important for Extract-Max(.)

The Heapify Operation

Constructing a Heap: Create-Heap(.)
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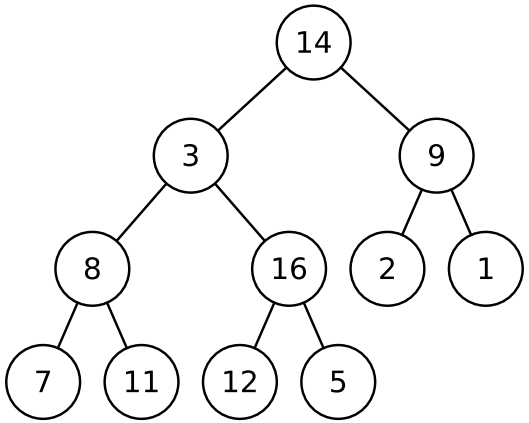

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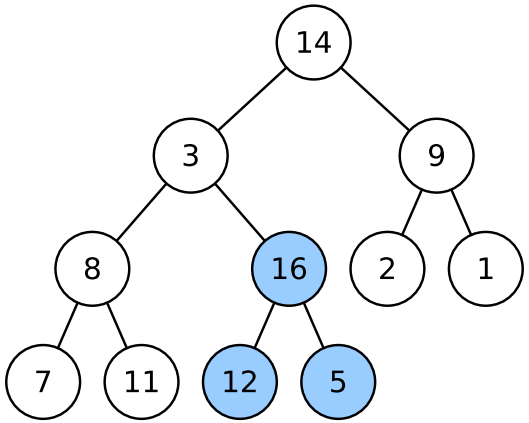

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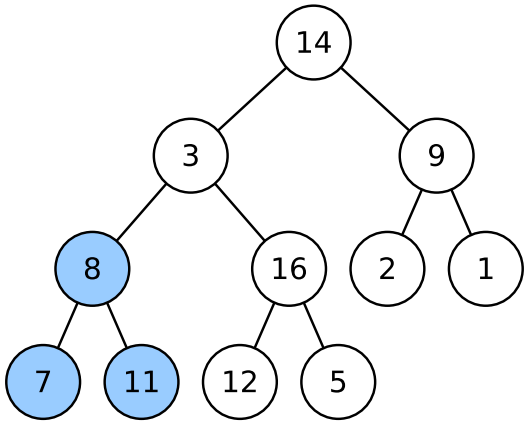

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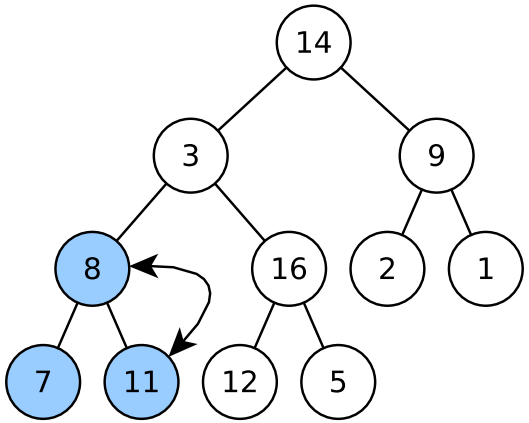

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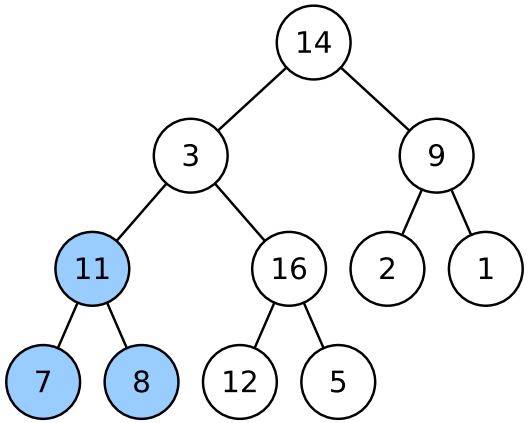

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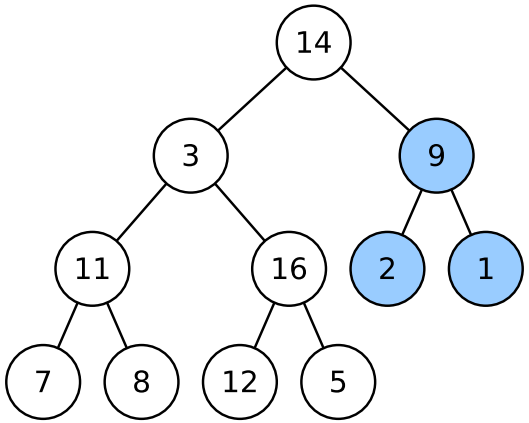

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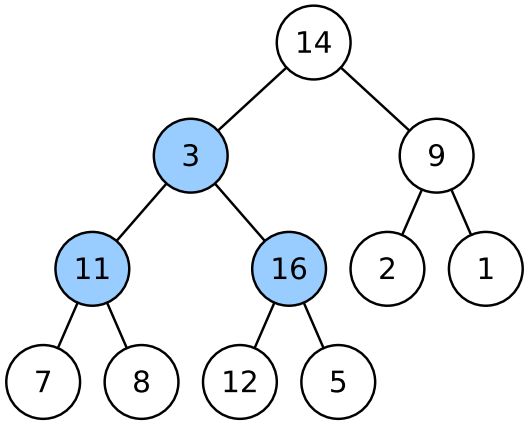

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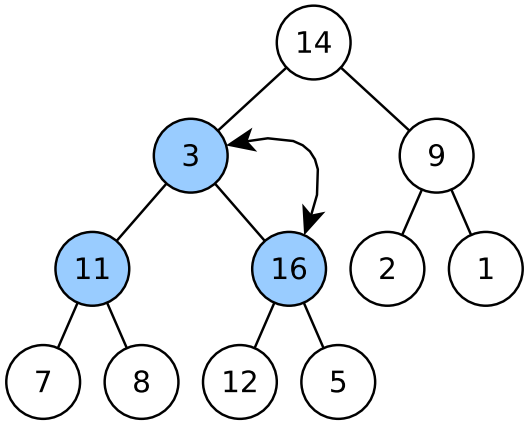

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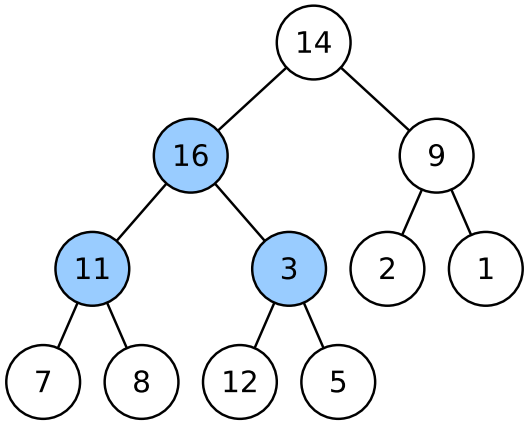

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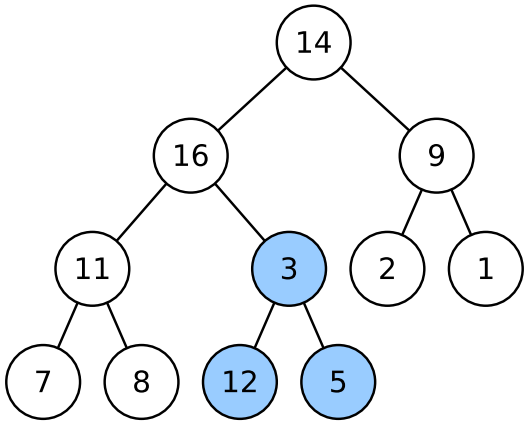

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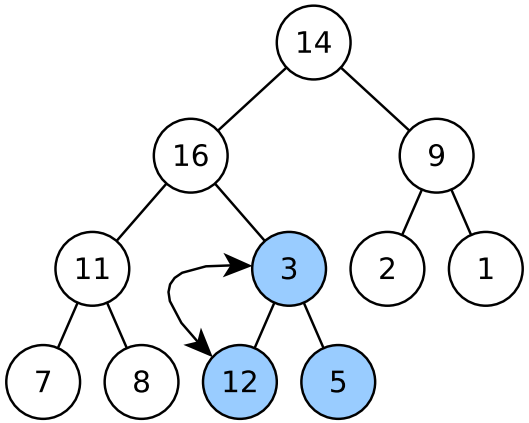

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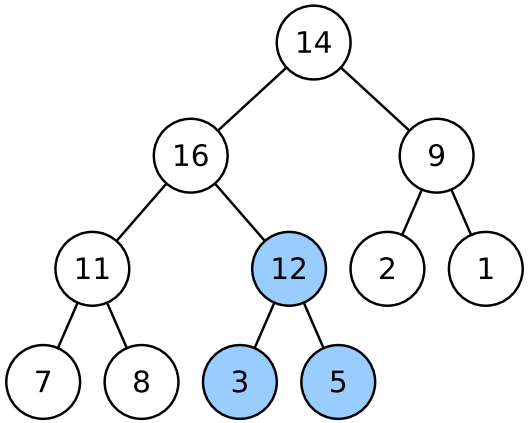

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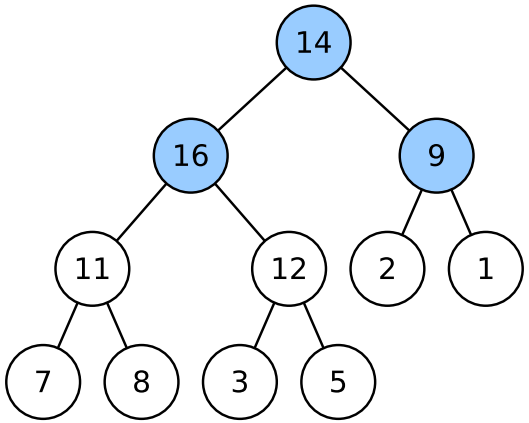

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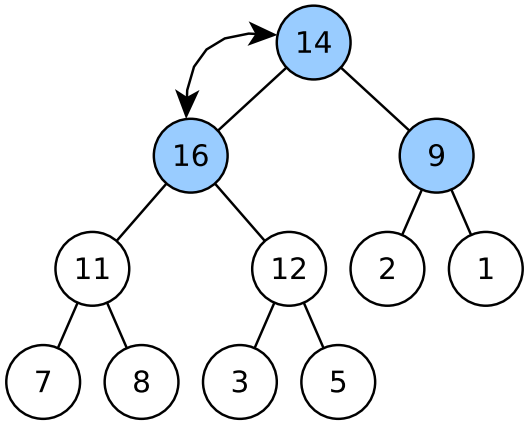

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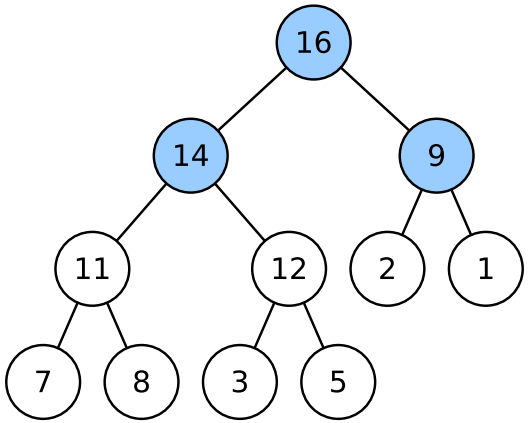

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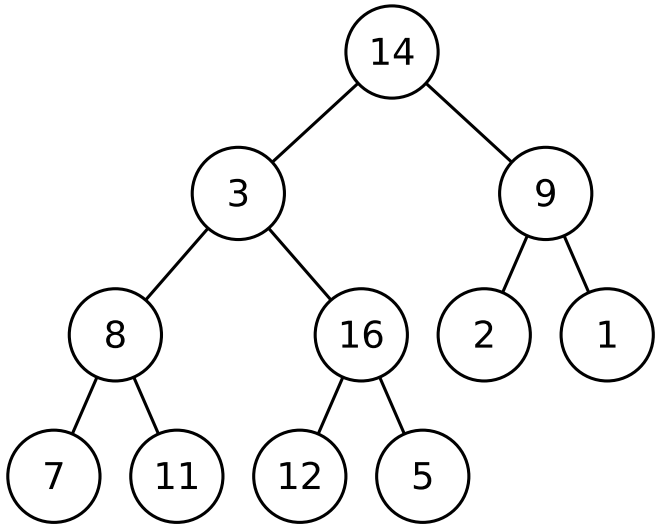

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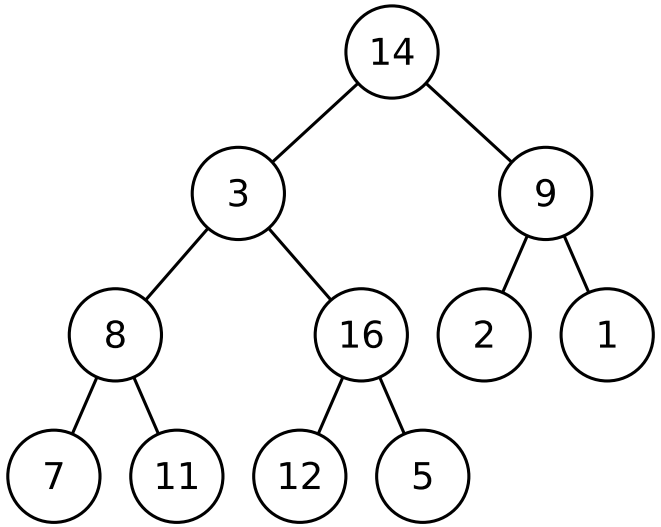

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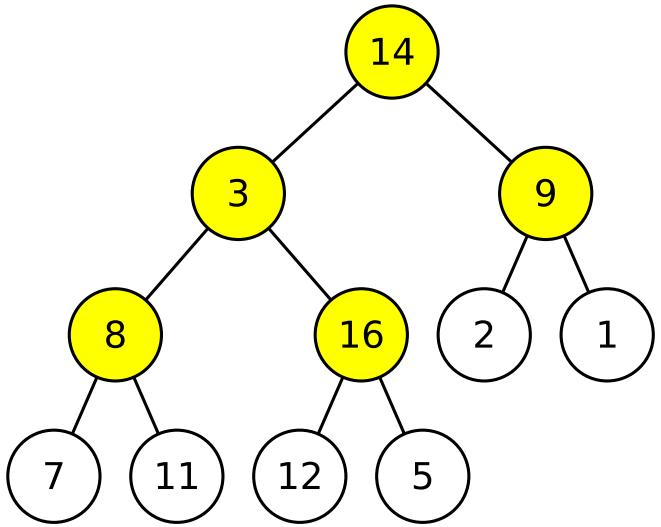

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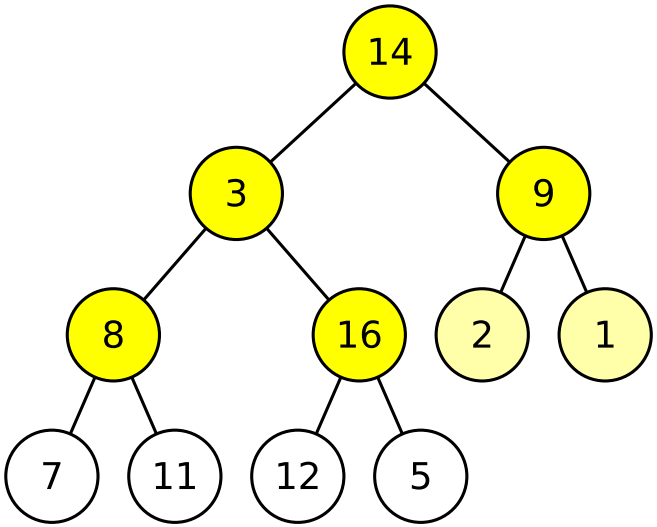
- These nodes are contained in a perfect binary tree



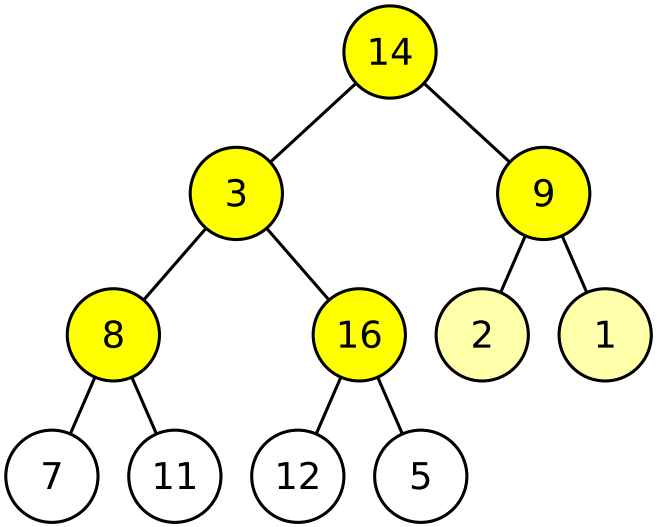
Improved Analysis of Heap Construction

More Precise Analysis of the Heap Construction Step

- Heapify(x): $O(\text{depth of subtree rooted at } x) = O(\log n)$
- **Observe:** Most nodes close to the "bottom" in a complete binary tree

Analysis:

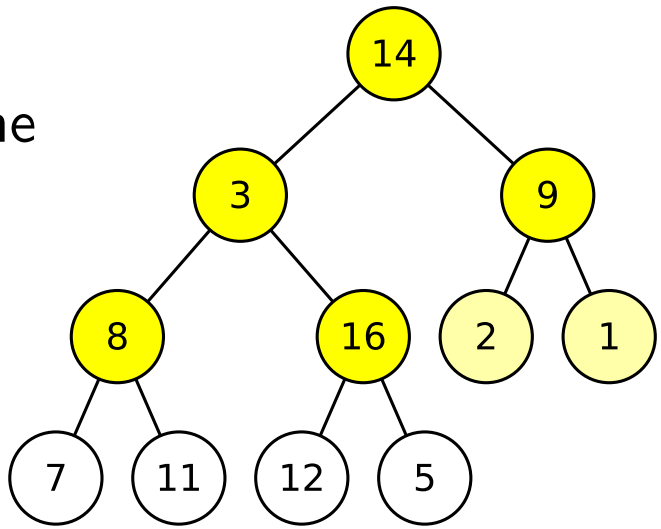
- Let i be the largest integer such that $n' := 2^i - 1$ and $n' < n$
- There are at most n' internal nodes (candidates for Heapify())
- These nodes are contained in a perfect binary tree
- This tree has i levels



Improved Analysis of Heap Construction

Analysis

We sum over all relevant levels, count the number of nodes per level, and multiply with the depth of their subtrees:



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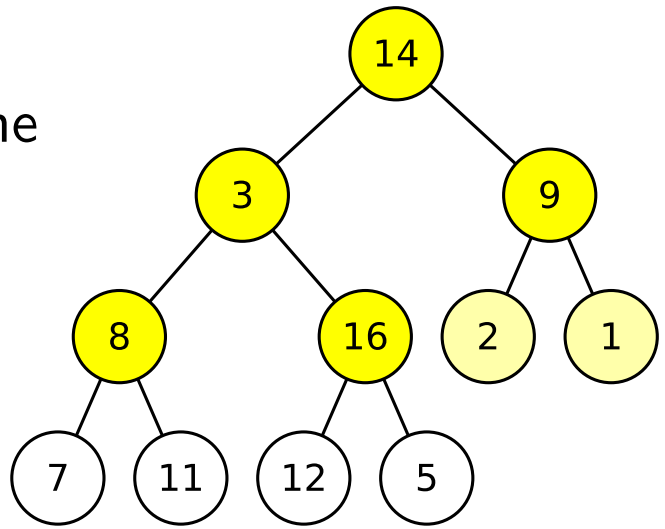

$$\text{Runtime} \quad = \quad \sum_{j=1}^{i} \# \text{ nodes at level } (i-j+1) \cdot \text{depth of subtree} \cdot O(1)$$

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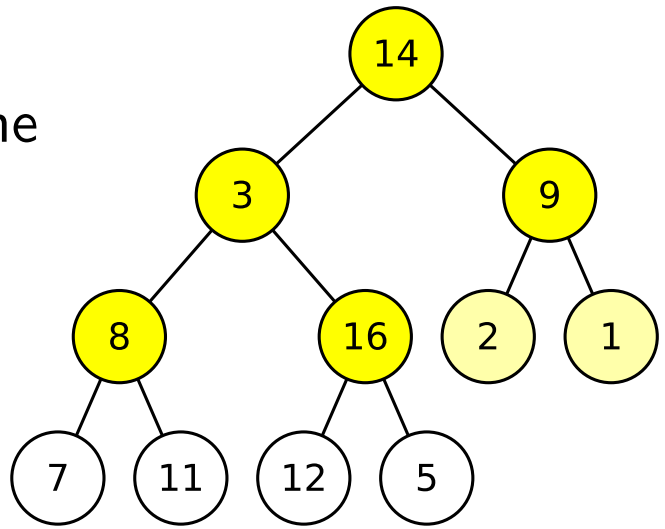

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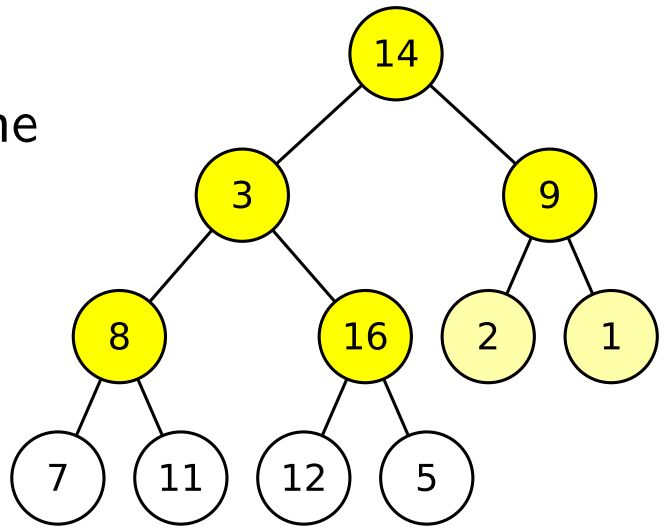

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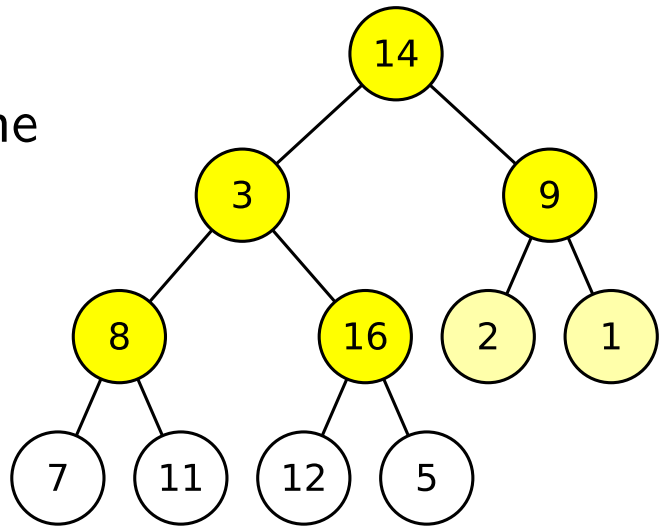

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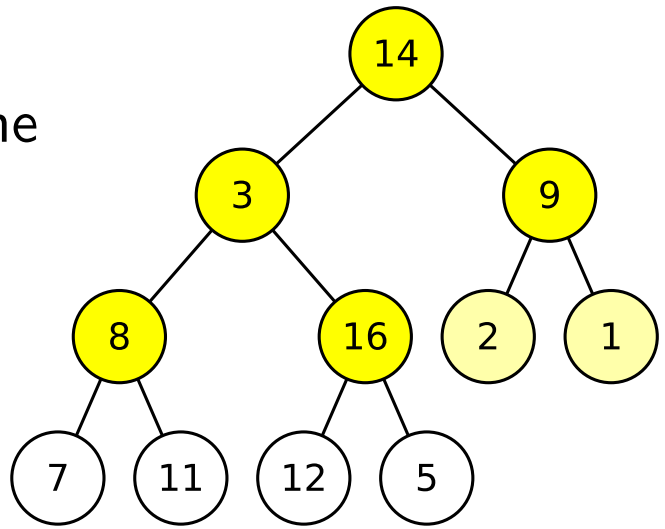

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Improved Analysis of Heap Construction

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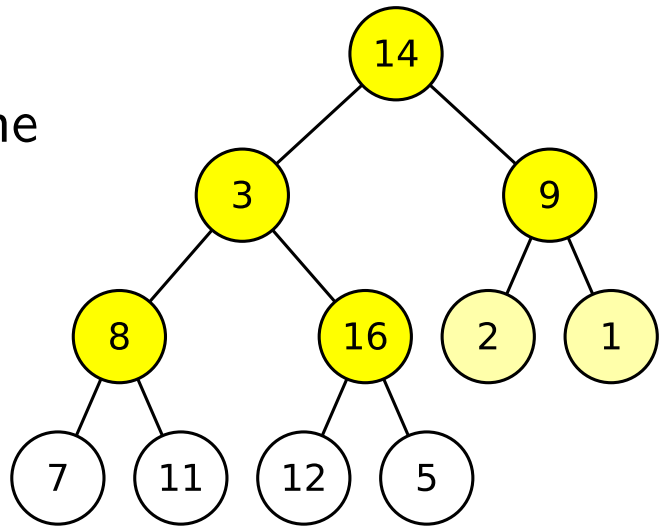

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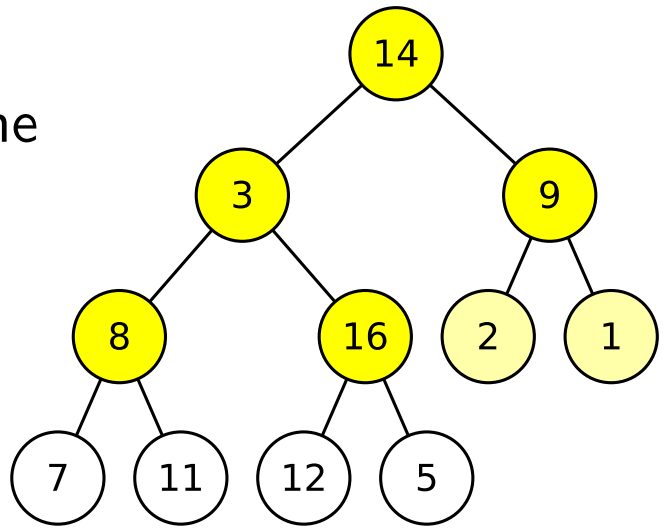

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\end{aligned}$$

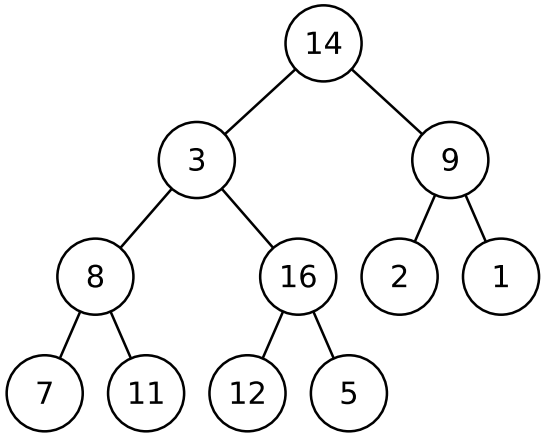
using $\sum_{j=1}^{i} \frac{j}{2^j} = O(1)$ (see trick from linear/binary search lecture).

The Complete Algorithm

Putting Everything Together

14	3	9	8	16	2	1	7	11	12	5

1. Create-Heap()
2. Repeat n times:
   1. Swap root with last element
   2. Decrease size of heap by 1
   3. Heapify(root)



The Complete Algorithm

Putting Everything Together

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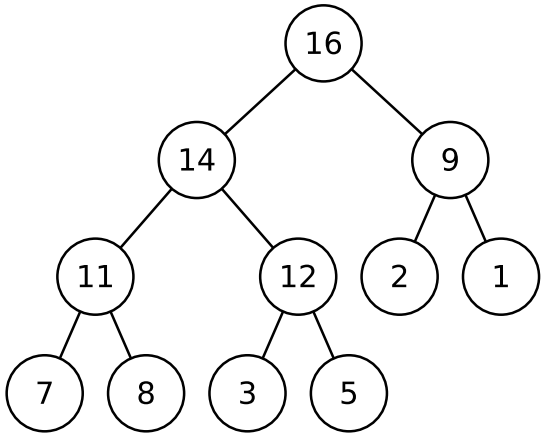

The Complete Algorithm

Putting Everything Together

5	14	9	11	12	2	1	7	8	3	16

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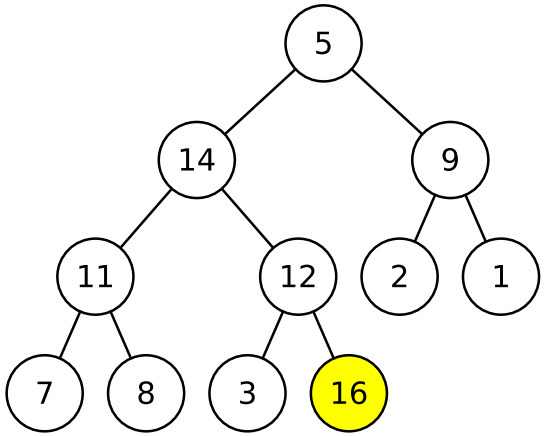

The Complete Algorithm

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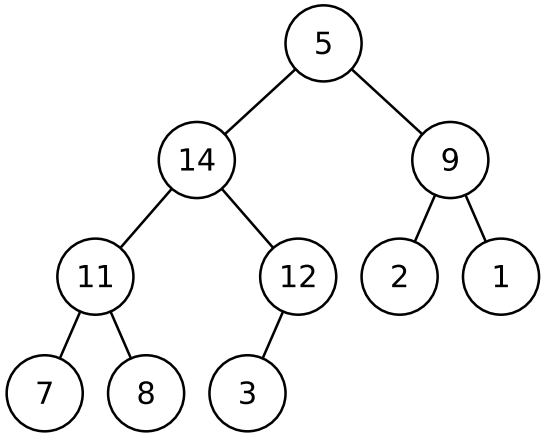

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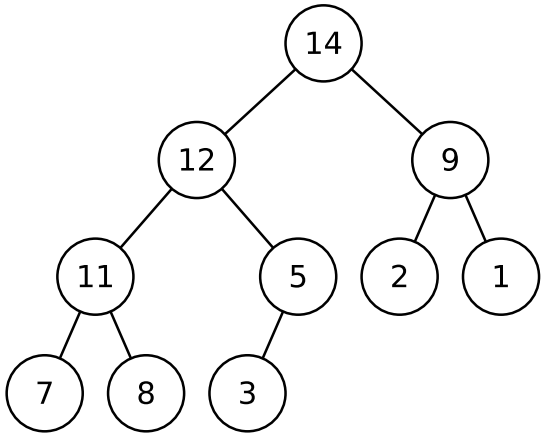

The Complete Algorithm

Putting Everything Together

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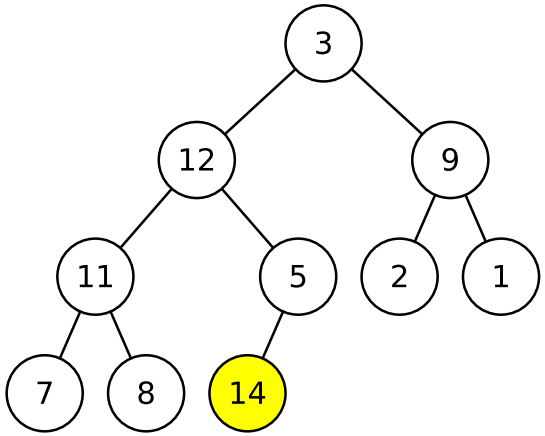

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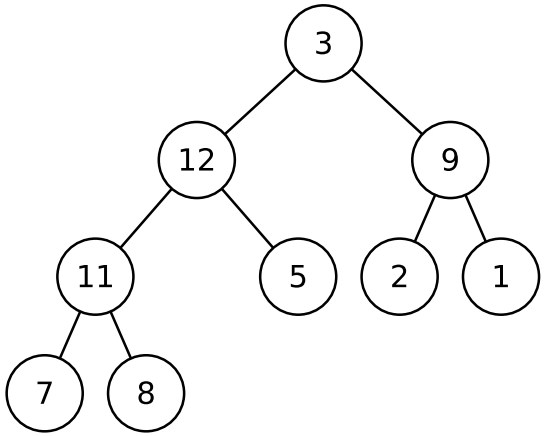

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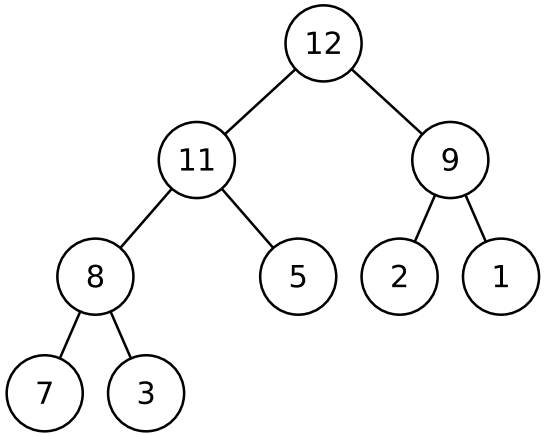

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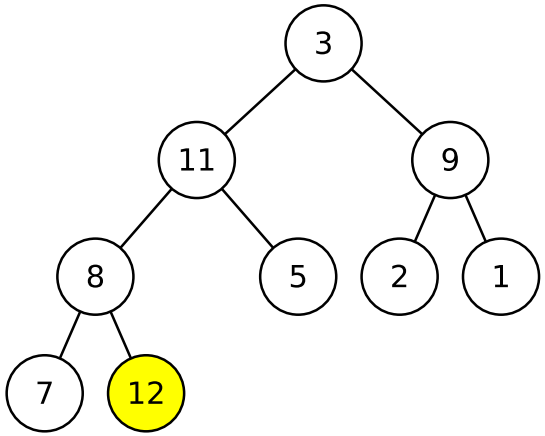

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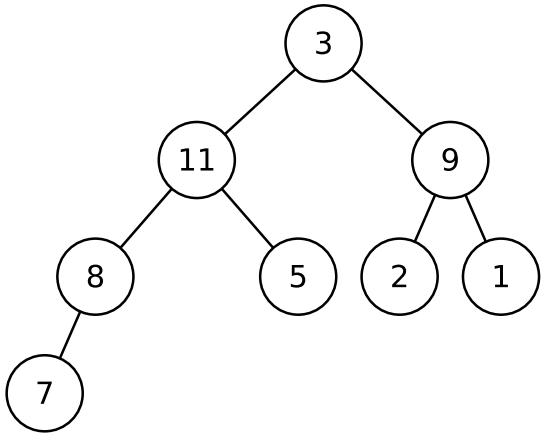

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The Complete Algorithm

Putting Everything Together

1	2	3	5	7	8	9	11	12	14	16

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The Complete Algorithm

Putting Everything Together

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1. Create-Heap() $O(n)$
2. Repeat n times:
   1. Swap root with last element $O(1)$
   2. Decrease size of heap by 1 $O(1)$
   3. Heapify(root) $O(\log n)$

Runtime: $O(n \log n)$

Heapsort is Not Stable

Example:

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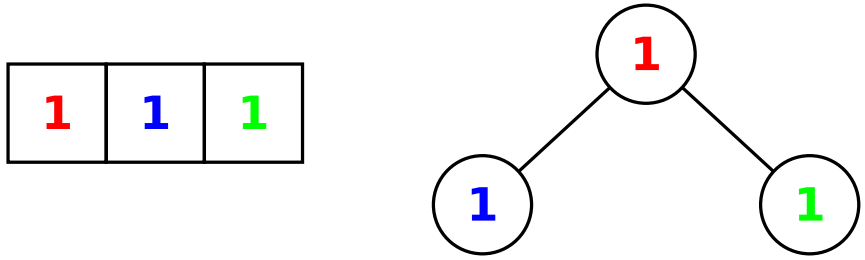

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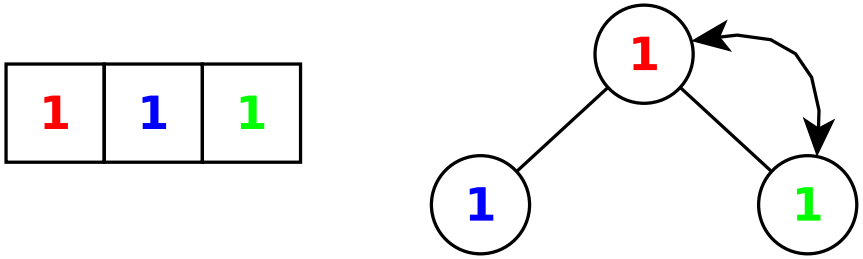

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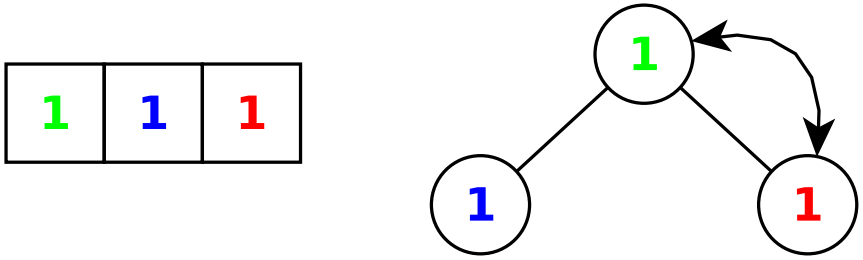

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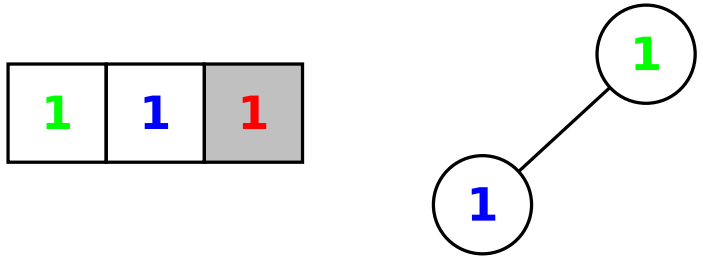

1 is moved from left to the right past 1 and 1

Heap-sort not stable